Abstract

The game of baseball is often decided by one or two runs, so base runners' decision to steal can have a high impact on winning or losing. This project uses SMT baseball data to study and analyze stolen base attempts from first base to second base. The data was filtered from ball-events, lineups, and play-positions then further narrowed down to plays involving a runner on first base, second base, short stop, and second baseman. Using Python and machine learning models, this project tested whether player movement features could help classify a steal attempt as successful or failed. The final model found that defensive movement features realted to defensive distance covered per step, velocity, acceleration, and jerk are the highest indicators for a steal prediction. Runner movement provided very high accuracy, so those results were concluded to be data leakage and not used in the final model. Throw-time analysis showed that catcher throws to the second baseman averaged 2.42 seconds with a 70% steal success rate, while throws to the shortstop averaged 2.41 seconds with a 62.75% steal success rate. Overall, the project found that the defensive movement helped predict stealing success.